Object Detection for Contraband Identification using Faster R-CNN on Full-Body X-ray Scans

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Introduction

Security screening in correctional facilities relies on manual inspection of full-body X-ray scans, which is slow and error-prone. Automated detection is challenging due to low grayscale contrast, overlapping structures, strong class imbalance, and very small contraband items that often occupy less than 1% of the image area. This project investigates whether preprocessing and data augmentation improve Faster R-CNN performance on grayscale X-ray imagery for contraband detection.

Research Questions

- Does $\log_2$ preprocessing improve detection accuracy in low-contrast X-ray images?
- Which augmentation strategies benefit or harm small-object detection?
- What preprocessing–augmentation setup yields the best overall performance?

Abstract

Security screening in correctional facilities relies on manual inspection of full-body X-ray scans, which is slow and error-prone. This work adapts Faster R-CNN for contraband detection in grayscale X-ray imagery and systematically evaluates the impact of $\log_2$ preprocessing and data augmentation. Experiments on 451 annotated scans show that minimal preprocessing without augmentation achieves the highest F1-scores, while contrast-enhancing and blurring augmentations significantly degrade small-object detection

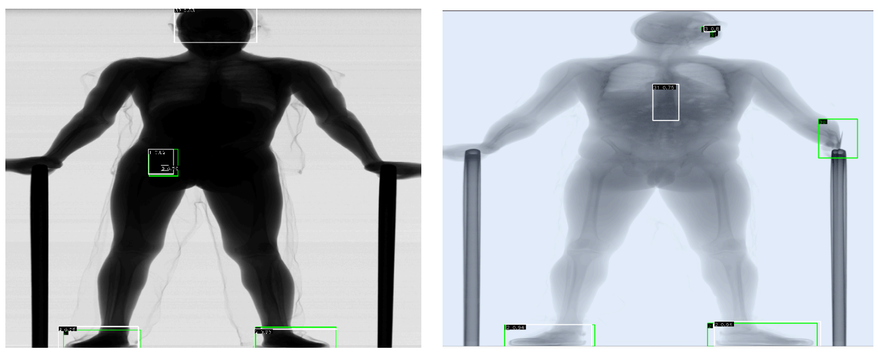

Figure 1. Raw Image VS Log 2 Filter with bounding boxes

Materials and Methods

Dataset

- Proprietary THEIA security X-ray dataset
- 451 full-body grayscale scans
- 33 object classes with strong imbalance
- Split: 50% train / 30% validation / 20% test (class-stratified)

Model

- Faster R-CNN (ResNet-50 + FPN)
- ImageNet pretrained backbone
- All layers trainable

Preprocessing

- Grayscale → 3-channel RGB
- Optional $\log_2(1 + I)$ intensity compression

Data Augmentation (training only)

- None (baseline), Gamma correction, Emboss, GaussianBlur and CLAHE

Evaluation

- Precision, Recall, F1-score at IoU = 0.5

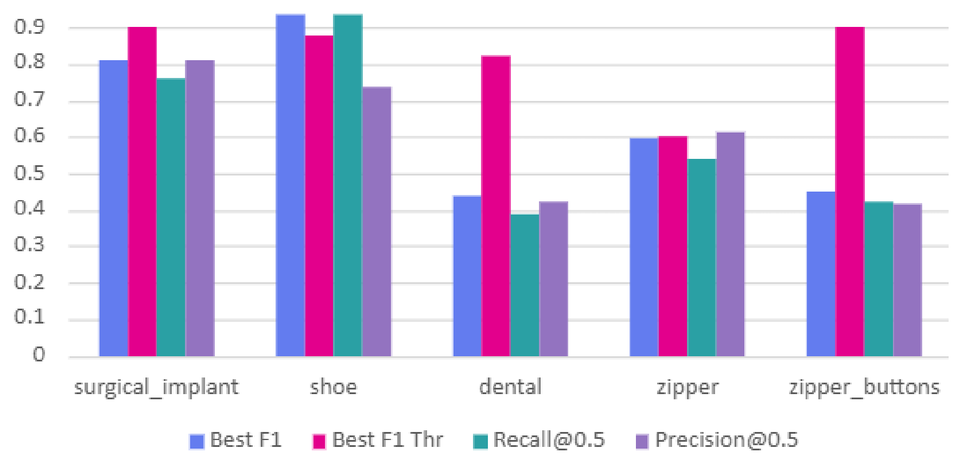

Figure 2 :Baseline Faster R-CNN without $\log_2$ filtering or augmentation achieves the highest overall F1-score.

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Experiments

Two main setups to isolate $\log_2$ and augmentation effects on Faster R-CNN:

Experiment 1: With $\log_2$ filter (enhances subtle densities but risks artificial edges/noise – tested on all 5 augmentations).

Experiment 2: Without $\log_2$ filter (raw contrast robustness, tested on all 5 augmentations)

All augmentation strategies are evaluated under both setups. Validation uses clean preprocessing with early stopping based on validation loss.

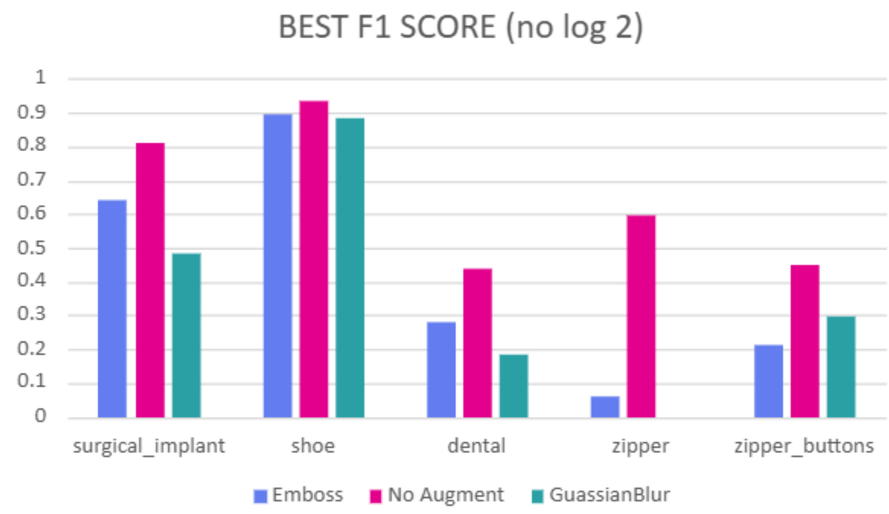

Figure 3: Performance drops sharply for small objects under heavy augmentation

Results

Overall detection performance is highest when Faster R-CNN is trained without $\log_2$ preprocessing and without data augmentation. Introducing $\log_2$ filtering or heavy augmentations consistently reduces recall and F1-score, particularly for small contraband objects. Contrast-enhancing methods (CLAHE) and blurring (GaussianBlur) performs worst, often suppressing fine object edges required for reliable region proposal generation.

Table 1: Overall performance summary

WITH LOG2	NO AUGMENT	GAMMA	EMBOSS	GAUSSIAN	CLAHE
BEST F1	0.445	0.2	0.2	0.128	0.202
RECALL	0.395	0.09	0.101	0.077	0.086
PRECISION	0.387	0.089	0.19	0.019	0.119
WITHOUT LOG 2	**NO AUGMENT**	**GAMMA**	**EMBOSS**	**GAUSSIAN**	**CLAHE**
BEST F1	0.503	0.277	0.239	0.24	0.25
RECALL	0.435	0.118	0.127	0.093	0.105
PRECISION	0.468	0.143	0.177	0.247	0.171

Table 2: Per-class F1-score (Top 5 classes)

CLASS	NO AUG(no log2)	BEST AUG(EMBOSS no log2)	WORST AUG(GAUSSIAN log2)
Surgical_Implant	0.812	0.641	0.027
shoe	0.936	0.898	0.722
dental	0.437	0.282	0.078
zipper	0.595	0.059	0
Zipper_buttons	0.449	0.214	0.0255

Conclusions

- **Optimal for all experiments**: No Augment without $\log_2$ – precise F1 gains (e.g., +0.043 on surgical_implant) from avoiding noise/artifacts.
- **Augmentation insights**: GaussianBlur/CLAHE weakest in both experiments (F1 near 0 on zipper/dental due to edge loss/uneven patches); Gamma/Emboss moderate but increases false positives. This shows that more refined augmentations could have a positive impact on the performance, in contrast to heavy augmentations.
- **Why? Faster R-CNN RPN is sensitive to clean features** – heavy augmentations distort X-ray densities in all setups.
- **Limitations**: Dataset size (451 scans) and class imbalance limit rare item precision; all experiments confirm simpler > complex for this domain
- **Contribution**: This work provides evidence-based guidelines for preprocessing and augmentation in grayscale X-ray object detection, showing that simpler pipelines outperform contrast-heavy approaches in security screening contexts.

References

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